



UNIVERSITY OF COLOMBO, SRI LANKA

UNIVERSITY OF COLOMBO SCHOOL OF COMPUTING

Bachelor of Science in Computer Science

First Year Examination - Semester 1 - 2021

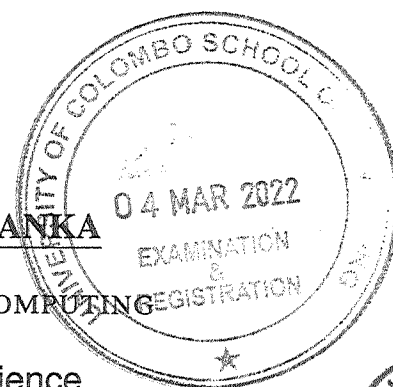
SCS 1202 — Programming using C — (Part A)

(2 hours for both parts)

ANSWER ALL QUESTIONS

Pages = 8

Questions = 2



237

To be completed by the candidate

Index Number

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Important Instructions

- This paper is in two parts, **Part A** and **Part B**. The candidate **must answer all questions**. The details are as follows:
Part A: 50 marks
(15 marks for Question 1 and 35 marks for Question 2)
Part B: 50 marks (25 marks for each question)
- Write your answers only in the answer boxes provided on this question paper.
- The candidate must answer the two parts separately.
- Do not tear off any part of this answer book. Under no circumstances may this book (or any part of this book), used or unused, be removed from the examination hall by a candidate.
- Questions appear on both sides of the paper. If a page is not printed, please inform the supervisor immediately.
- Any electronic device capable of storing and retrieving text including electronic dictionaries and mobile phones are **not allowed**.
- Please write your answers only in English.

To be completed by the examiners

1	
2	
Total	

1. (a) C is a *row-major* language. What does it mean? Explain with an example.

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[3 marks]

- (b) (i) The following program gave a *segmentation fault* during its execution. Suggest a solution for it.

```
#include <stdio.h>

int main() {
    int i = 20;
    int distance[10];

    for (int i=0; i<10; ++i)
        distance[i] = 0;

    distance[i-1] = 1;
}
```

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[2 marks]

(ii) What is the problem with the following *if* statement?

```
int j;  
/* ... */  
/* some code  
/* ... */  
if (j=4)  
    printf("j=%d\n", j);
```

.....
.....

[1 mark]

(c) Write down the output of the following C code.

```
#include <stdio.h>  
  
main() {  
    short I=13;  
    short J=11;  
    short N;  
    N = I&J;  
    printf("N=%d\n", N);  
    printf("Done\n");  
}
```

.....
.....

[3 marks]

(d) If the variable *c* is defined to be an *int* and if it is filled through the statement *c=getchar()* then

```
if (c >= '0' && c <= '9')
```

checks whether the character in *c* is a digit. Explain it.

.....
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.....

[3 marks]

- (e) A teacher asks you to write the code lines in the following code in order to print the elements of the matrix *C* row-by-row. i.e., each row of the matrix should be printed on one line with the column elements separated by tab spaces. The elements of the next line should be similarly printed on the next line and so on.

Fill the four lines of the code (indicated by -----) to fulfill the above need.

```
#define N 4    /* Matrix Size */

int C[N][N];

int main(int argc, char *argv[]) {
    int i,j;

    /* ... */
    /* some code that will fill the C matrix with values */
    /* ... */

    /* print C */
    #ifdef PRINT
    printf("Matrix C:\n");

    -----
    -----
    -----
    -----

    }
    #endif
}
```

[3 marks]

2. (a) A *prime number* is a number that is divisible only by itself and 1.

Fill the five blanks (indicated by -----) in the following code in order to find whether an input number is a *prime* or not.

```
#include <stdio.h>

int main() {

    int n, i, flag = -----; /* BLANK 1 */
    printf("Enter a positive integer: ");
    scanf("%d", &n);

    // 0 and 1 are not prime numbers
    // change flag to 1 for non-prime number

    if (n == 0 ----- n == 1) /* BLANK 2 */
        flag = 1;

    for (i = 2; i <= n / 2; ++i) {

        if (-----) { /* not a prime */ /* BLANK 3 */

            flag = 1;

            -----; /* BLANK 4 */
        }
    }

    // flag is 0 for prime numbers
    if (flag == 0)
        printf("%d is a prime number.", n);

    ----- /* BLANK 5 */

    printf("%d is not a prime number.", n);

    return 0;
}
```

[10 marks]

- (b) The following incomplete code is designed to use recursion to print the reverse of an input number. Fill the five blanks (indicated by -----) to complete it.

Note: Given below is an output from a program run:

Enter a number:4719

Reversed number is:9174

```
#include <stdio.h>

int reverse(int);

int ans=0,rem;

int main(){
    int num,reverseNo;

    printf("\nEnter a number:");
    scanf("%d",&num);

    /* call the function to reverse the number */

    reverseNo=-----; /* BLANK 1 */
    printf("\nReversed number is :%d\n",reverseNo);
    return 0;
}

int reverse(int -----){ /* BLANK 2 */
    if(num){
        -----=num%10; /* BLANK 3 */

        -----=ans*10+rem; /* BLANK 4 */

        reverse(-----); /* BLANK 5 */
    }
    else
        return ans;
}
```

[10 marks]

- (c) There are many sorting algorithms and *bubble sort* is one of them. It works as follows:

Consider $a[n]$ to be the array of numbers to be sorted.

First pass: Compare the first element with the next one (i.e., $a[0]$ and $a[1]$) and swap if $a[0] > a[1]$. Now compare $a[1]$ and $a[2]$ and swap if $a[1] > a[2]$. Repeat this process until the end of the array is reached. When this is done, the largest element is present at the end of the array. In this first pass, the array elements from 0 to $n-1$ are processed.

Second pass: Repeat the above work, but process array elements $a[0]$ to $a[n-2]$ because the last one, i.e., $a[n-1]$, is present at its correct position. After this pass, the largest two elements are present at the end of the array.

Similarly repeat the passes until the entire array is sorted.

- (i) An *incomplete bubble sort* program is given below.

Complete the code by filling the four blanks (indicated by -----).

```
/* Bubble sort code */

#include <stdio.h>

int main() {
    int array[100], n, i, j, temp;

    printf("Enter number of elements to sort:");
    scanf("%d", &n);

    printf("Enter %d integers\n", n);

    for (i = 0; i < n; i++)
        scanf("%d", &array[i]);

    for (i = 0; i < n - 1; i++)
    {
        for (j = 0; j < -----; j++) /* BLANK 1 */
        {
            if (array[j] > array[j+1])
            {
                temp = -----; /* BLANK 2 */

                ----- = array[j+1]; /* BLANK 3 */

                -----; /* BLANK 4 */
            }
        }
    }

    printf("Sorted list in ascending order:\n");

    for (i = 0; i < n; i++)
        printf("%d\n", array[i]);

    return 0;
}
```

[10 marks]

- (ii) Write down a problem with this program with regard to the number n and suggest a solution for it.

.....
.....
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.....
.....
.....

[5 marks]



University of Colombo, Sri Lanka

University of Colombo School of Computing
BACHELOR OF SCIENCE IN COMPUTER SCIENCE

First Year Examination — Semester I— 2021

SCS1202 — Programming using C - Part B

(Two (2) Hours)

Answer ALL questions

Number of Pages = 8

Number of Questions = 2

248

To be completed by the candidate

Index Number

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Important Instructions to candidates:

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- Note that questions appear on both sides of the paper. If a page is not printed, please inform the supervisor immediately.
- Write your index number on each and every page of the question paper.
- The duration of the paper is **Two (2)** hours for **both parts A & B**.
- This paper, **part B** has **2** questions on **8** pages.
- Answer **all** the questions in this **part B**
- Each question carries exactly **25** marks.
- Write your answers on the space provided on this question paper.
- Any electronic device capable of storing and retrieving text including electronic dictionaries and mobile phones are **not allowed**.

To be completed by the examiners

1	
2	
Total	

Index Number

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1. (a). Only ONE answer is correct in the following 10 MCQs. Underline the **best suite answer** among the given options.

[2 x 12 = 24 marks]

- i. What among the following is true?

- A. Pointers are to store the values of complex data structures.
- B. Pointers are type of arrays.
- C. Pointers increase the complexity of the program.
- D. Pointers increase the execution speed.
- E. Pointers use to passing data by value.

- ii. `int *x = &y;`

according to the above line of code written in C, the variable x is a/an,

- A. integer variable.
- B. memory address.
- C. complex data type.
- D. variable to hold the value of y.
- E. given name for a memory location.

- iii. Which of the following is true regarding the variable y of the line of code given in the question (ii) above?

- A. y is an array.
- B. y is a pointer.
- C. y is an integer.
- D. y is a memory address.
- E. y is a constant.

- iv. A function q that accepts a pointer to a character as argument and returns an integer can be declared as,

- A. `int q(char ch)`
- B. `int* q(char ch)`
- C. `int q(char &ch)`
- D. `int* q(char* ch)`
- E. `int q(char* ch)`

- v. If `sizeof(x)` returns the value 1, the variable type of x can be a

- A. struct.
- B. double.
- C. float.
- D. integer.
- E. char.

Index Number

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vi. If *ptr* is integer pointer variable and the value of *vpr* is 0062FE14, the value of *ptr++* is;

- A. 0062FE15
- B. 0062FE10
- C. 0062FE18
- D. 0062FE1C
- E. 0062FE20

vii. Consider following piece of code written in C.

```
1 int age[7] = {46, 56, 20, 85, 12, 21, 84};  
2 int *ptr = age;  
3 printf("\%d, ", *(ptr+3));  
4 printf("\%d", *ptr+3);
```

What would be the values printed in the screen when we run this program?

- A. 20, 23
- B. 85, 88
- C. 49, 85
- D. 20, 49
- E. 85, 49

viii. What is true among the following regarding the *Unions* in C.

- A. Members of a union should be in same type.
- B. Members of an union can have the same name.
- C. Union is a simple variable type.
- D. We can use only one union member at a time.
- E. Size of an union is equal to the sum of the sizes of its member variables.

ix. What is true among the following regarding the *Structures* in C.

- A. Members of a structure must be in different types.
- B. Size of a structure is a constant.
- C. Initial values of its members can be assigned in struct definition.
- D. A structure can contain an instance of itself.
- E. Structure variables cannot be compared.

x. Consider following piece of code written in C.

```
1 enum arr {a=10, b=8, c=6, d, e=2};  
2 printf("%d", d);
```

According to the above code, the output will be printed in the screen is;

- A. 3
- B. 4
- C. 5
- D. 6
- E. 7

Index Number

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xi. Consider following piece of code written in C.

```
1 void main() {  
2   enum weekdays {mon, tue, wed, thu, fri};  
3   enum weekdays day;  
4   for (day=mon; day<=fri; day++) {  
5     printf("%d, ", day);  
6   }  
7 }
```

According to the above code, the output will be printed in the screen is;

- A. mon, tue, wed, thu, fri,
- B. 0, 1, 2, 3, 4,
- C. 1, 2, 3, 4, 5,
- D. a compilation error
- E. a runtime error

xii. Consider following piece of code written in C.

```
1 union abc {  
2   int figure;  
3   char letter;  
4 };  
5  
6 void main() {  
7   union abc myUnion;  
8   myUnion.figure = 65;  
9   myUnion.letter = 'B';  
10  printf("%d, ", myUnion.figure);  
11  printf("%c", myUnion.letter);  
12 }
```

According to the above code, the output will be printed in the screen is;

- A. 65, A
- B. 65, B
- C. 66, A
- D. 66, B
- E. a compilation error

(b). You are suppose to store students' names in an array, defined (in C) as follows.

```
char *studentName [10];
```

What is the length of a name you can store in this array?

[1 mark]

Index Number

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2. (a). The following program (written in C) convert the number you enter into a secret code and a secret value as defined by the function named `SecretCode()`.

```
1 int SecretCode(int x) {  
2   return (x*x)/2; }  
3  
4 void main() {  
5   int num;  
6   printf("Enter your number: ");  
7   scanf("%d", &num);  
8   int sc = SecretCode(num);  
9   printf("Your Secret Code: %d\n", sc);  
10  printf("Your Secret value: %d\n", sc*2);  
11  printf("done!");  
12 }
```

- i. What is the output of the above program, if you enter 5 as the input?

[2 marks]

--

- ii. Re-write the above program again to do the same, but passing the arguments to the function by reference.

[7 marks]

--

Index Number

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- (b). The following program (written in C) supposed to print 65 66 67 68 E in the console when it runs.

```
1 struct element {
2 int a, b, c; };
3
4 struct unit {
5 int x;
6 struct element y;
7 char z; };
8
9 // print_element() function signature goes here...
10
11 void main() {
12 struct unit u = {65, {66, 67, 68}, 69};
13 printf("%d ", u.x);
14 print_element (1);
15 printf(2, 3);
16 }
17
18 // implementation of the print_element() function goes here
19
20 }
```

- i. write down the piece of code that should come to the number (1), (2) and (3)?

[3 marks]

(1).

(2).

(3).

- ii. Give the implementation of the print_element () function of the above program.

[4 marks]

Index Number

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(c). The following program (written in C) compiles with no errors.

MyTxt.txt is a file located in the same folder and the initial content of it is "UCSC".

```
1 void main() {  
2  
3 FILE *fp = fopen("MyTxt.txt", "r");  
4 char ch=getc(fp);  
5 ch=getc(fp);  
6  
7 while(!feof(fp)) {  
8 ch=getc(fp);  
9 printf("%c", ch);  
10 }  
11  
12 if(feof(fp))  
13 printf("\nOK!");  
14 fclose(fp);  
15  
16 fopen("MyTxt.txt", "w");  
17 fprintf(fp, "SCS1202");  
18 fprintf(fp, "C Programming");  
19 printf("done!");  
20 fclose(fp);  
21 }
```

i. What would display in the console when this program runs?

[3 marks]

--

ii. What is the content of the MyTxt.txt file after it runs?

[2 marks]

--

--	--	--	--	--	--	--	--

- [4 marks]



UNIVERSITY OF COLOMBO, SRI LANKA



UNIVERSITY OF COLOMBO SCHOOL OF COMPUTING

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

First Year Examination – Semester I – 2021/2022

SCS1201 – Data Structures and Algorithms I

TWO (2) HOURS (For Part A & B)

PART (A)



058

To be completed by the candidate

Examination Index No:

Important Instructions to candidates:

1. The medium of instruction and question is **English**.
2. Write your answers in English
3. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
4. Note that questions appear on both sides of the paper. If a page is not printed, please inform the supervisor immediately.
5. Write your index number on each and every page of the Question paper.
6. Part A has 03 questions in 14 pages.
7. Answer **ALL** questions.
8. **Part A** of the paper will total to **60 marks (20 marks for each question)** and **Part B** of the paper will total to **40** marks.
9. This paper consists of two parts, Part A (Question No 1, Question No 2 and Question 3) and Part B (Question No 4 only) and submit both Parts separately.
10. Any electronic device capable of storing and retrieving text including electronic dictionaries and mobile phones are **not allowed**.
11. **Non-Programmable** calculators are **allowed**.

For Examiner's use only

For Examiner's use only	
Question No	Marks
1	
2	
3	
4	
Total	

Index No: ...

Part A

Question 1

(a) Consider the following algorithms:

- Tower of Hanoi
- Fibonacci number series
- Tree traversal

One student proposed the following conclusion after carefully considering the implementation of the above algorithms.

“The above algorithms can be implemented using only recursive approach”

As a skilled student of implementation of Data Structures and Algorithms, discuss the validity of the above conclusion.

[4 Marks]

(b) The quality of an algorithm depends on the following characteristics.

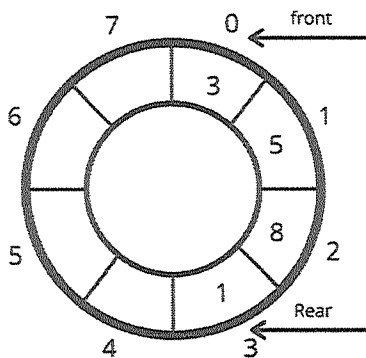
- Unambiguous.
- Input
- Output
- Finiteness
- Feasibility
- Independent

Discuss how one would incorporate the above characteristics when designing a quality algorithm?

[6 Marks]

Index No: ...

(c) Consider the following circular queue with initial values.



Initials values are :

Front=0

Rear=3

Count=4

(i) Write Front, Rear, and Count values after two (02) data elements are deleted from the Queue.

[2 Marks]

Index No: ...

Front =
Rear =
Count =

- (ii) Write Values for Front, Rear, and Count after insertion of following data elements to the queue obtained from Part C(i) above.

Data elements are : 6,2,7,9,4

[2 Marks]

Front =
Rear =
Count =

- (d) Consider that one processor is shared between 5 terminals (you may assume that the terminals do not have a processing power). Suppose the terminals submit their jobs to the processor according to the following terminal ID order.
- A
 - B
 - C
 - D
 - E

After submitting their jobs to the processor, the operating system will generate a priority value, and the priority values with terminal ID's are shown in the following table.

Terminal ID	A	B	C	D	E
Priority Value	8	12	3	5	22

- (i) Construct a Max-Heap based on the initial submission order. You may apply the top-down fixing approach if there are any violations of the heap property.

[3 Marks]

Index No: ...

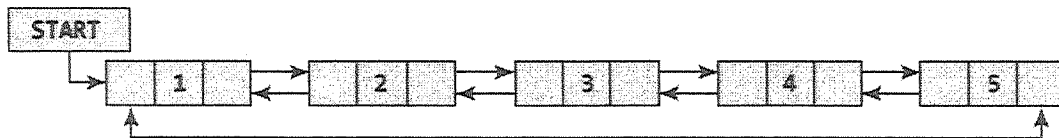
- (ii) Illustrate how to sort the jobs in different terminals according to the increasing order of priority values without using an additional array.

[3 Marks]

Index No: ...

Question 2

(a) Consider the following circular doubly linked list.



(i) Write a Pseudocode or C code to find a given node from the above-linked list.

[3 Marks]

(i)

Index No: ...

- (ii) Write a Pseudocode or C code to delete the first Node (i.e. data element =1) from the above linked list.

[3 Marks]

- (b) Using a suitable example(s), show the validity of the following statements.

Note: h is the height of the Binary tree

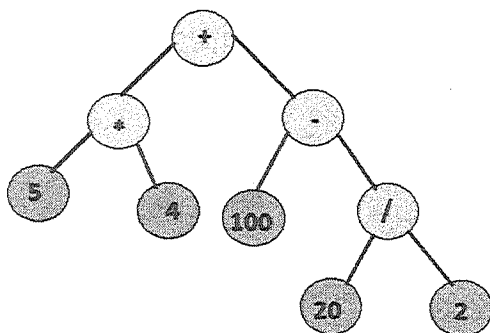
- (i) The total number of leaf nodes in a complete binary tree is at least $h+1$ and at most 2^h
(ii) The total number of nodes in a complete binary tree is at least $2h+1$ and at most $2^{h+1} - 1$

[2x2=4 Marks]

Continued.

Index No: ...

(c) Consider the following expression tree.



What is the order of nodes of the trees if traversed in the following sequence?

- (i) Pre-order
- (ii) In-order
- (iii) Post-order

[4 Marks]

Index No: ...

(i)
(ii)
(iii)

(d) The following algorithm is used to insert new nodes into Binary Search Tree.

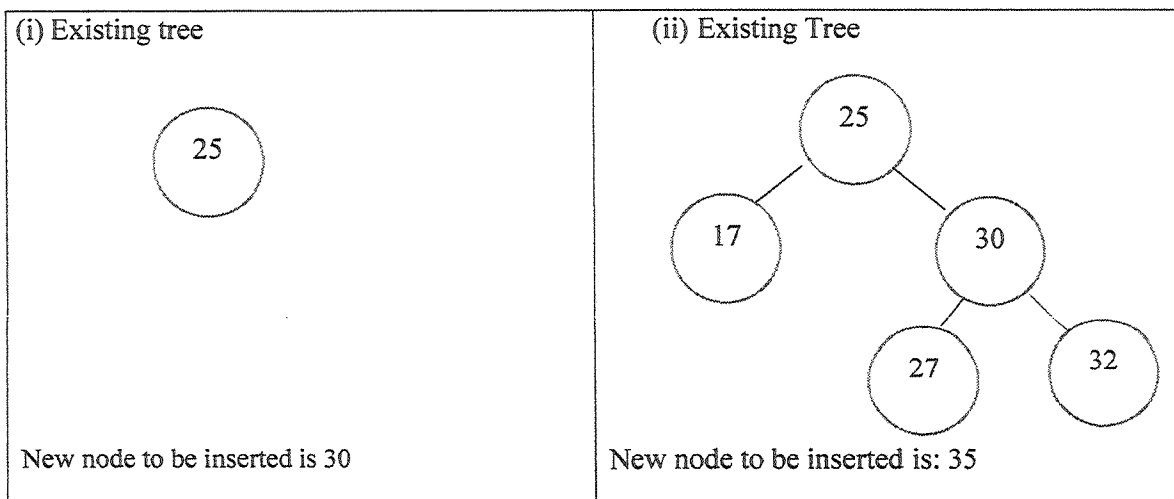
Input: binary search tree T, node v, element e

Output:

```
Insert(T, v, e){  
    if(T.isLeaf(v)){  
        if(v.element() >= e)  
            Insert element e as v's left child  
        else  
            Insert element e as v's right child  
    } else {  
        if(v.element() >= e)  
            Insert (T, T.leftChild(v), e)  
        else  
            Insert (T, T.rightChild(v), e)  
    }  
}
```

Note: The content of the new node is denoted by e.

Illustrate how the above pseudocode algorithm works when inserting the given nodes to the existing tree.



[3x2=6 Marks]

Index No: ...

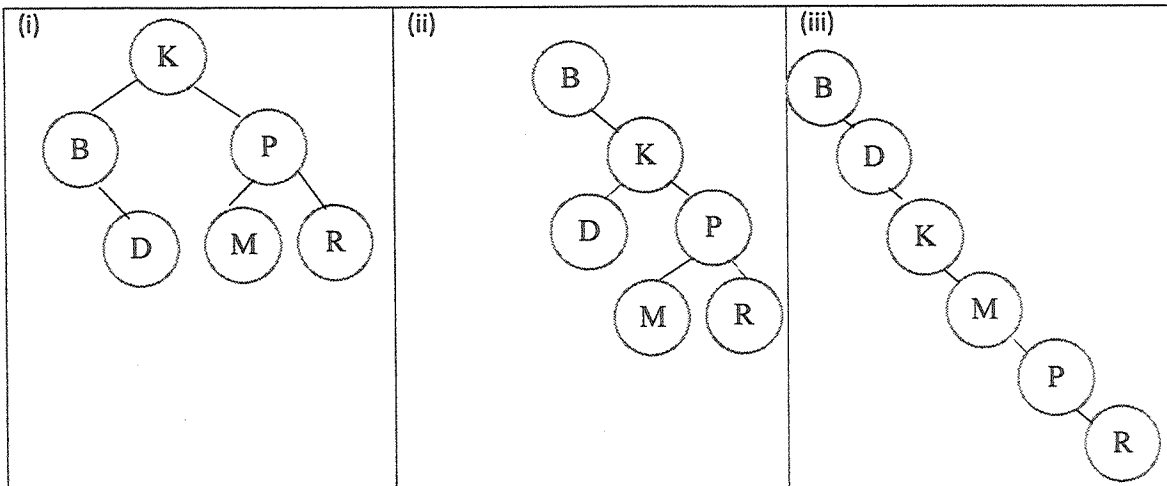
Question 3

(a) Define Self-Balancing Binary Search Trees.

[3 Marks]

Index No: ...

- (b) Determine whether the following trees are balanced or not. If a tree is not balanced, discuss why it is not balanced.



(2x3=6 Marks)

- (c) Day–Stout–Warren algorithm (DSW) algorithm for balancing a tree has two significant steps as follows:

Step 1: First, create a backbone or vine

Step 2: Transform the backbone/vine to a well-balanced tree.

Note: For the two questions C(i) and C(ii) given below you should provide all the intermediate steps when creating the backbone (part C(i)) and transforming it into a well-balanced tree (part C(ii)).

- (i) Consider the following algorithm, which illustrates Step 1 of the DSW Algorithm.

CreateBackbone(root,n)

Tmp=root

While (Tmp != null)

If Tmp has a left child

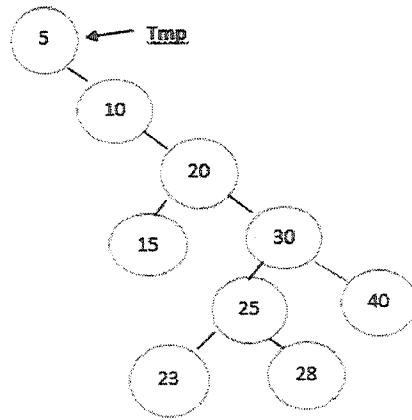
Rotate this child about the temp

Set Tmp to the Child which Just Become parent

Else Set Tmp to its right child

Index No: ...

Transform the following binary search tree into a backbone/vine using the above algorithm.



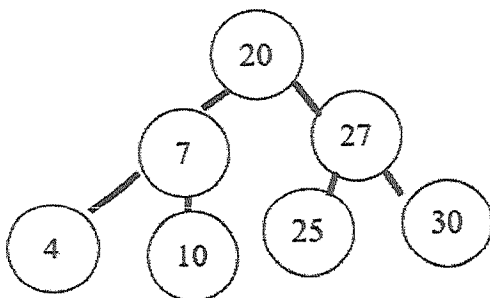
[4 Marks]

Index No: ...

(ii) Using Suitable rotations, transform the backbone/vine created in Part (c) (i) above into a well-balanced tree.

[4 Marks]

(d) Consider the following AVL tree.



Insert the following nodes to the above tree.

23, 26, 21

It was noted that the AVL properties are violated after inserting the new nodes. Discuss how the AVL tree can be restored using suitable rotations.

[3 Marks]

Index No: ...



UNIVERSITY OF COLOMBO, SRI LANKA

UNIVERSITY OF COLOMBO SCHOOL OF COMPUTING

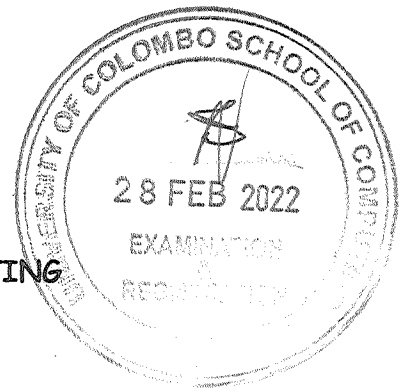
BACHELOR OF SCIENCE IN COMPUTER SCIENCE

Academic Year 2020/2021 – First Year Examination – Semester I – 2021

SCS1201 – Data Structures and Algorithms I

TWO (2) HOURS

PART B



089

To be completed by the candidate

Examination Index No:

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7. **PART B** has **01** question across **4** pages.
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only**

Question No	Marks
4	
Total	

Index No.

Question 04

- (a) Consider an application that requires sorting an array of numbers. The application environment has memory constraints, and it is found that a **memory write is a costly operation**. What is the most suitable sorting algorithm to use in this scenario? Explain your answer with algorithmic reasoning, time, and memory complexity.

[15 Marks]

Index No.

- (b) Consider a scenario where a set of names consisting of the first and last names are to be sorted. The sorting scheme requires the names to be sorted first by the first name and subsequently by the last name. If sorting intended to sort the names by the first name **AND** the last name, what aspects need to be considered when choosing a sorting algorithm? Explain your answer with algorithmic reasoning.

[15 Marks]

Index No.

- (c) Providing algorithmic reasoning agree or disagree with the following statement. Explain your reasons.

“There are instances that insertion sort performs better than the merge sort for an unsorted array.”
[05 Marks]

- (d) Explain what is meant by in-place sorting and name two algorithms for in-place sorting.

[05 Marks]
